

TANNU SINGH

Frontend Engineer · React.js · Typescript · Node.js · Generative AI
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EXPERIENCE

Tech Mahindra · Software Engineer

Sep 2024 – Present

- Architected a 20-component React UI library with Material UI and a global custom CSS-in-JS theme, adopted across multiple Nuix products — eliminating inconsistent UI patterns company-wide and enabling zero-rework integration for downstream teams.
- Delivered a production-grade calendar component with full responsiveness and MUI theme compliance, shipped on deadline after resolving 100+ PR review comments from Sydney-based client engineers — demonstrating ownership under real cross-timezone pressure.
- Became the team's MUI theming authority within the first sprint; designed the global theme architecture that unblocked the broader product release and standardized styling decisions across engineering.
- Leveraged AI-First Frontend Engineering System to accelerate component scaffolding, debug CSS-in-JS conflicts, and systematically review PR patterns — reducing iteration cycles and increasing personal throughput on high-comment PRs.

Exavalu · React.js Engineering Intern

Feb 2024 – Jul 2024

- Independently resolved 25% of the client ticket backlog in 5 months — targeting long-standing React bugs and UI regressions that had blocked product teams for weeks.
- Refactored components to eliminate unnecessary re-renders, improving rendering efficiency and contributing to on-time delivery across 3 simultaneous client projects.

PROJECT

AI System Design Explorer

- Built an AI-powered System Design Explorer that generates Staff Engineer-level architecture breakdowns for products like Uber, Netflix, WhatsApp, and Airbnb from simple natural language inputs.
- Engineered a modular React and TypeScript architecture with reusable components, custom hooks, and AI-driven workflows to analyze requirements, services, infrastructure, scalability, and engineering trade-offs.
- Designed multiple learning modes including Architecture Explorer, Deep-Dive Analysis, Interview Mode, and Revision Mode to help developers understand and prepare for real-world system design interviews.
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- Developed an AI orchestration layer with prompt engineering, report generation, history management, and structured architecture rendering, producing recommendations for databases, caching, messaging systems, security, fault tolerance, and scaling strategies.

Crypto Tracker Dashboard · React · Chakra UI · CoinGecko API

- Built a live cryptocurrency dashboard polling CoinGecko API on a configurable interval, tracking 100+ coins with real-time price updates, 24h % change, and market cap rankings — entirely client-side, no backend.
- Engineered a multi-coin comparison view for side-by-side price trend visualization of up to 5 coins using Recharts with custom tooltips and responsive breakpoints.
- Optimized re-render cycles by 35% using useMemo and selective state updates, maintaining smooth UI during high-frequency polling without throttling the update rate.
- Implemented persistent user preferences (watchlist, currency, theme) via localStorage, ensuring full state rehydration on every visit.

SKILLS

Frontend Engineering: React.js, Vue.js, JavaScript, TypeScript, Next.js, Tailwind CSS, Material UI, HTML/CSS, jQuery

Generative AI: OpenAI API Integration, Prompt Engineering, RAG, LLM Application Development, Engineering Workflow Automation

State & Architecture: Redux Toolkit, Zustand, TanStack React Query, Design Systems, Frontend Architecture

Backend & APIs: Node.js, Express.js, GraphQL, Rest APIs, JWT Authentication, Middleware, MongoDB, Mongoose, MySQL, SQL, Redis

Tools, Testing & Cloud: Jest, Docker, AWS, Git, GitHub, GitLab, Postman, Agile/Scrum

EDUCATION

Haldia Institute of Technology · B.Tech, Electronics & Communication Engineering

Oct 2020 – Jul 2024 · CGPA: 8.27 / 10

CERTIFICATION

AWS Certified Cloud Practitioner

June 2026 – June 2029